

Technology Stack

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, Bootstrap 5, and Flask (Jinja2) templates	Provides a simple, responsive form for analysts and customers to enter applicant details and view instant approval/rejection results, without needing a separate front-end build pipeline.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3 with Flask	Same language as the ML training pipeline (scikit-learn, XGBoost), so the trained model loads directly into the app; Flask is lightweight and fast to develop routes/APIs with for a prediction-focused service.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV dataset for model training; MySQL for storing applications and prediction logs	The historical credit-approval dataset is processed from CSV during training; MySQL provides reliable relational storage for new applicant submissions, prediction results, and audit records used in compliance reporting.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	IBM Cloud with IBM Watson Machine Learning; Docker for containerizing the Flask app	IBM Watson ML hosts the trained model for scalable, real-time predictions as required by the project; Docker keeps the runtime environment consistent across development and deployment.

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5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	GitHub	Enables seamless collaboration across the five-member team through branches and pull request reviews, and integrates directly with the IBM Cloud deployment workflow.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	IBM Watson Machine Learning REST API	Lets the Flask web app send applicant data to the cloud-hosted model and receive an approval/rejection prediction in real time, enabling scalable production use beyond local inference.